

Roy Liu

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EDUCATION

University of Texas at Austin

Bachelor of Science in Computer Science, GPA: 3.89/4.0

Austin, TX

Expected May 2028

EXPERIENCE

Quantum Hardware Intern

Logiqal, Inc.

May 2026 – Aug 2026

Monmouth Junction, NJ

- Redesigned a Solarflare ef_vi kernel-bypass receive path for FPGA waveform acquisition, increasing receive throughput by 18×.
- Built a distributed NATS control platform for 40+ laboratory instruments supporting neutral-atom quantum computing experiments, with real-time monitoring and device control.
- Developed closed-loop AOM calibration achieving < 1% waveform error in < 0.2 ms, validated with RFSoc hardware-in-the-loop testing.
- Built a PostgreSQL/TimescaleDB telemetry backend, reducing storage by ~6× through compression and average query latency by ~2×.

Undergraduate Research Assistant

GAMMA Lab, University of Texas at Austin

Oct 2025 – Present

Austin, TX

- First author of “Observation Geometry for Uncertainty-Aware Hamiltonian Inference and Experimental Design in Quantum Magnets,” [arXiv:2608.10350](https://arxiv.org/abs/2608.10350).
- Developed a FiLM-modulated SIREN surrogate for neutron-scattering simulations, reducing model evaluation from days to seconds with > 0.9 Pearson correlation.
- Built a 7.2-billion-point training dataset on TACC using sharded HDF5 files and striped Lustre I/O.
- Built a Bayesian experimental-design pipeline that adaptively selects measurements to maximize information gain under experimental constraints.

Simulation & Validation

Longhorn Racing Combustion

Sep 2025 – Present

Austin, TX

- Built a Raspberry Pi telemetry system decoding CAN data from 20+ vehicle sensors and streaming it to Oracle Cloud for live monitoring and validation.
- Developed a limited-slip differential torque model in Julia using differential-algebraic equations and clutch friction models.
- Extended the team’s Python vehicle model with anti-roll bar dynamics, inverse motion-ratio calculations, and interpolated damper force curves.

PROJECTS

Quantum Pulse Optimization | Python, SciPy, GRAPE

2026

- Built a GRAPE-based optimizer for single- and two-qubit Rydberg gate pulses using gradient-based numerical optimization.
- Modeled AOM bandwidth, phase transients, and fiber dispersion as differentiable operators to account for hardware distortion during optimization.

SKILLS

Languages: Python, C, C++, Java, Julia, SQL

ML & Scientific: PyTorch, NumPy, pandas, SciPy, scikit-learn, XGBoost, HDF5

Systems & Infrastructure: Linux, Docker, NATS, Git, PostgreSQL, TimescaleDB, Grafana, FPGA/DMA